

Tool Specification Document

Attribute	Value
Document Type	Technical Tool Specification
Prepared For	Engineering Implementation Team
Version	1.0
Status	Draft for Implementation Review

1. Project Overview

The AI Productivity Agent is a large language model (LLM)-powered assistant designed to manage tasks and meetings through natural language interaction. The system is built on the following technology stack:

Component	Role
FastAPI	Exposes the agent as a REST/HTTP service and handles request/response lifecycle.
LangGraph	Orchestrates the agentic control flow as a stateful graph (planning, tool selection, execution, and response generation nodes).
LangChain	Provides the tool-calling abstraction layer, prompt templates, and message history management.
Ollama (Local LLM)	Serves as the local inference engine that performs natural language understanding, intent resolution, and tool argument extraction.
SQLite	Provides persistent, transactional storage for tasks, meetings, and derived artifacts (e.g., action items).

1.1 Design Principle: No Rule-Based Intent Classification

A foundational architectural constraint of this project is that the LLM alone is responsible for determining user intent and selecting the correct tool. The system does not use:

- Keyword matching (e.g., if "delete" in message)
- Regex-based intent extraction
- Rule-based or decision-tree classifiers
- Hard-coded command parsing

Instead, all tools are registered with the LLM as callable functions (via LangChain's tool-calling / function-calling interface), each with a strict JSON Schema describing its purpose, arguments, and constraints. The LLM reads the user's natural language message, reasons over the available tool schemas, and emits a structured tool call (tool name plus arguments), which the LangGraph runtime then executes deterministically. This document specifies every tool at the level of contract (schema, validation, behavior, and error handling) required for an engineer to implement the tool layer without inspecting existing source code.

1.2 Common Conventions Used Throughout This Document

- All timestamps use ISO 8601 format (YYYY-MM-DDTHH:MM:SS) in the system's local timezone unless otherwise noted.
- All tool outputs follow a consistent envelope structure: { "status": "success | error", "data": { }, "error": null }
- All entity identifiers (task_id, meeting_id) are UUID v4 strings generated at creation time.
- Enum fields are case-sensitive and restricted to the values listed.

- Every Write operation is executed within a SQLite transaction; on failure, the transaction is rolled back and an error object is returned.
- Tools marked Approval Required: Yes must not execute their underlying state-changing action until an explicit confirmation signal is received from the human user via the ApprovalTool workflow (see Section 13).

2. Tool: CreateTask

Tool Name	CreateTask
Purpose	Creates a new task record in the system with a title, optional description, due date, priority, and status. Invoked when the user expresses intent to add, log, or remember an actionable item.
Input Schema	<pre>{ "type": "object", "properties": { "title": { "type": "string", "minLength": 1, "maxLength": 200 }, "description": { "type": "string", "maxLength": 2000 }, "due_date": { "type": "string", "format": "date-time" }, "priority": { "type": "string", "enum": ["low", "medium", "high", "urgent"] }, "tags": { "type": "array", "items": { "type": "string" } } }, "required": ["title"] }</pre>
Required Fields	title
Optional Fields	description, due_date, priority (defaults to "medium"), tags
Output Schema	<pre>{ "status": "success", "data": { "task_id": "uuid", "title": "string", "description": "string null", "due_date": "string null", "priority": "string", "status": "pending", "tags": ["string"] } }</pre>

	<pre>], "created_at": "datetime" }, "error": null } </pre>
Read/Write Operation	Write
Approval Required	No
Possible Errors	INVALID_TITLE — title missing or empty; TITLE_TOO_LONG — exceeds 200 characters; INVALID_DATE_FORMAT — due_date not ISO 8601; INVALID_PRIORITY — value not in enum; DB_WRITE_FAILURE — SQLite transaction failed.
Example Call	<pre> { "tool": "CreateTask", "arguments": { "title": "Prepare Q3 budget report", "due_date": "2026-07-25T17:00:00", "priority": "high", "tags": ["finance", "reporting"] } } </pre>
Example Result	<pre> { "status": "success", "data": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "title": "Prepare Q3 budget report", "description": null, "due_date": "2026-07-25T17:00:00", "priority": "high", "status": "pending", "tags": ["finance", "reporting"], "created_at": "2026-07-18T09:12:44" }, "error": null } </pre>

3. Tool: UpdateTask

Tool Name	UpdateTask
Purpose	Modifies one or more fields of an existing task (title, due date, priority, description, tags) without altering its identity or creation metadata.
Input Schema	<pre> { "type": "object", "properties": { "task_id": { "type": "string", "format": "uuid" }, "title": { "type": "string", </pre>

	<pre> "minLength": 1, "maxLength": 200 }, "description": { "type": "string", "maxLength": 2000 }, "due_date": { "type": "string", "format": "date-time" }, "priority": { "type": "string", "enum": ["low", "medium", "high", "urgent"] }, "tags": { "type": "array", "items": { "type": "string" } } }, "required": ["task_id"] } </pre>
Required Fields	task_id
Optional Fields	title, description, due_date, priority, tags (at least one optional field must be supplied)
Output Schema	<pre> { "status": "success", "data": { "task_id": "uuid", "updated_fields": ["string"], "task": { "task_id": "uuid", "title": "string", "description": "string null", "due_date": "string null", "priority": "string", "status": "string", "tags": ["string"], "updated_at": "datetime" } }, "error": null } </pre>
Read/Write Operation	Write
Approval Required	No

Possible Errors	TASK_NOT_FOUND — no task with given task_id; NO_FIELDS_PROVIDED — request contains only task_id; INVALID_DATE_FORMAT; INVALID_PRIORITY; DB_WRITE_FAILURE.
Example Call	<pre>{ "tool": "UpdateTask", "arguments": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "priority": "urgent", "due_date": "2026-07-22T17:00:00" } }</pre>
Example Result	<pre>{ "status": "success", "data": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "updated_fields": ["priority", "due_date"], "task": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "title": "Prepare Q3 budget report", "description": null, "due_date": "2026-07-22T17:00:00", "priority": "urgent", "status": "pending", "tags": ["finance", "reporting"], "updated_at": "2026-07-18T09:20:11" } }, "error": null }</pre>

4. Tool: DeleteTask

Tool Name	DeleteTask
Purpose	Permanently removes a task record from the system. This is a destructive, irreversible operation and therefore requires human approval prior to execution.
Input Schema	<pre>{ "type": "object", "properties": { "task_id": { "type": "string", "format": "uuid" }, "reason": { "type": "string", "maxLength": 300 } }, "required": ["task_id"] }</pre>
Required Fields	task_id

Optional Fields	reason
Output Schema	<pre>{ "status": "success", "data": { "task_id": "uuid", "deleted": true, "deleted_at": "datetime" }, "error": null }</pre>
Read/Write Operation	Write
Approval Required	Yes — must be routed through ApprovalTool before deletion is committed.
Possible Errors	TASK_NOT_FOUND; APPROVAL_PENDING — action awaiting user confirmation; APPROVAL_REJECTED — user declined the deletion; DB_WRITE_FAILURE.
Example Call	<pre>{ "tool": "DeleteTask", "arguments": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "reason": "Duplicate task entry" } }</pre>
Example Result	<pre>{ "status": "success", "data": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "deleted": true, "deleted_at": "2026-07-18T09:25:03" }, "error": null }</pre>

5. Tool: CompleteTask

Tool Name	CompleteTask
Purpose	Marks an existing task as completed, recording a completion timestamp. Used when the user reports finishing an action item.
Input Schema	<pre>{ "type": "object", "properties": { "task_id": { "type": "string", "format": "uuid" }, "completion_note": { "type": "string", "maxLength": 300 } }, "required": ["task_id"] }</pre>

Required Fields	task_id
Optional Fields	completion_note
Output Schema	<pre>{ "status": "success", "data": { "task_id": "uuid", "status": "completed", "completed_at": "datetime", "completion_note": "string null" }, "error": null }</pre>
Read/Write Operation	Write
Approval Required	No
Possible Errors	TASK_NOT_FOUND; TASK_ALREADY_COMPLETED — task status already completed; DB_WRITE_FAILURE.
Example Call	<pre>{ "tool": "CompleteTask", "arguments": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "completion_note": "Submitted to finance team" } }</pre>
Example Result	<pre>{ "status": "success", "data": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "status": "completed", "completed_at": "2026-07-18T09:30:55", "completion_note": "Submitted to finance team" }, "error": null }</pre>

6. Tool: ListTasks

Tool Name	ListTasks
Purpose	Retrieves a filtered and/or paginated list of tasks. Used for general queries such as "show me my tasks" or "what's due this week."
Input Schema	<pre>{ "type": "object", "properties": { "status": { "type": "string", "enum": ["pending", "completed", "all"] }, "priority": {</pre>

	<pre> "type": "string", "enum": ["low", "medium", "high", "urgent"] }, "due_before": { "type": "string", "format": "date-time" }, "due_after": { "type": "string", "format": "date-time" }, "limit": { "type": "integer", "minimum": 1, "maximum": 100 }, "offset": { "type": "integer", "minimum": 0 } }, "required": [] } </pre>
Required Fields	None
Optional Fields	status (defaults to "pending"), priority, due_before, due_after, limit (defaults to 20), offset (defaults to 0)
Output Schema	<pre> { "status": "success", "data": { "tasks": [{ "task_id": "uuid", "title": "string", "priority": "string", "status": "string", "due_date": "string null" }], "total_count": "integer", "limit": "integer", "offset": "integer" }, "error": null } </pre>
Read/Write Operation	Read
Approval Required	No
Possible Errors	INVALID_DATE_RANGE — due_after later than due_before; INVALID_PAGINATION — limit/offset out of range; DB_READ_FAILURE.
Example Call	<pre> { "tool": "ListTasks", "arguments": { "status": "pending", </pre>

	<pre> "due_before": "2026-07-25T23:59:59", "limit": 10 } } </pre>
Example Result	<pre> { "status": "success", "data": { "tasks": [{ "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "title": "Prepare Q3 budget report", "priority": "urgent", "status": "pending", "due_date": "2026-07-22T17:00:00" }], "total_count": 1, "limit": 10, "offset": 0 }, "error": null } </pre>

7. Tool: SearchTasks

Tool Name	SearchTasks
Purpose	Performs free-text search across task titles, descriptions, and tags. Used for queries like "find the task about the budget report."
Input Schema	<pre> { "type": "object", "properties": { "query": { "type": "string", "minLength": 1, "maxLength": 200 }, "limit": { "type": "integer", "minimum": 1, "maximum": 50 } }, "required": ["query"] } </pre>
Required Fields	query
Optional Fields	limit (defaults to 10)
Output Schema	<pre> { "status": "success", "data": { "results": [{ "task_id": "uuid", "title": "string", "relevance_score": "number", "status": "string" }] } } </pre>

	<pre> }], "result_count": "integer" }, "error": null } </pre>
Read/Write Operation	Read
Approval Required	No
Possible Errors	EMPTY_QUERY; QUERY_TOO_LONG; NO_RESULTS_FOUND (returned as a successful empty result, not a hard error); DB_READ_FAILURE.
Example Call	<pre> { "tool": "SearchTasks", "arguments": { "query": "budget report", "limit": 5 } } </pre>
Example Result	<pre> { "status": "success", "data": { "results": [{ "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "title": "Prepare Q3 budget report", "relevance_score": 0.94, "status": "pending" }], "result_count": 1 }, "error": null } </pre>

8. Tool: CreateMeeting

Tool Name	CreateMeeting
Purpose	Schedules a new meeting with a title, start/end time, participants, and optional location or agenda notes.
Input Schema	<pre> { "type": "object", "properties": { "title": { "type": "string", "minLength": 1, "maxLength": 200 }, "start_time": { "type": "string", "format": "date-time" }, "end_time": { "type": "string", "format": "date-time" } } } </pre>

	<pre> "participants": { "type": "array", "items": { "type": "string" } }, "location": { "type": "string", "maxLength": 200 }, "agenda": { "type": "string", "maxLength": 2000 } }, "required": ["title", "start_time", "end_time"] } </pre>
Required Fields	title, start_time, end_time
Optional Fields	participants, location, agenda
Output Schema	<pre> { "status": "success", "data": { "meeting_id": "uuid", "title": "string", "start_time": "datetime", "end_time": "datetime", "participants": ["string"], "location": "string null", "agenda": "string null", "created_at": "datetime" }, "error": null } </pre>
Read/Write Operation	Write
Approval Required	No
Possible Errors	INVALID_TIME_RANGE — end_time earlier than or equal to start_time; INVALID_DATE_FORMAT; MEETING_CONFLICT — overlapping meeting detected for the same participants (warning-level, non-blocking unless configured strict); DB_WRITE_FAILURE.
Example Call	<pre> { "tool": "CreateMeeting", "arguments": { "title": "Sprint Planning", "start_time": "2026-07-21T10:00:00", "end_time": "2026-07-21T11:00:00", "participants": ["alice@company.com", "bob@company.com"], "location": "Conference Room B" } } </pre>

	}
Example Result	<pre>{ "status": "success", "data": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "title": "Sprint Planning", "start_time": "2026-07-21T10:00:00", "end_time": "2026-07-21T11:00:00", "participants": ["alice@company.com", "bob@company.com"], "location": "Conference Room B", "agenda": null, "created_at": "2026-07-18T09:40:12" }, "error": null }</pre>

9. Tool: UpdateMeeting

Tool Name	UpdateMeeting
Purpose	Modifies details of an existing meeting, such as rescheduling the time, changing participants, or updating the agenda.
Input Schema	<pre>{ "type": "object", "properties": { "meeting_id": { "type": "string", "format": "uuid" }, "title": { "type": "string", "minLength": 1, "maxLength": 200 }, "start_time": { "type": "string", "format": "date-time" }, "end_time": { "type": "string", "format": "date-time" }, "participants": { "type": "array", "items": { "type": "string" } }, "location": { "type": "string", "maxLength": 200 }, "agenda": { "type": "string", "maxLength": 2000 } }, "required": ["meeting_id"] }</pre>

	<pre>] } </pre>
Required Fields	meeting_id
Optional Fields	title, start_time, end_time, participants, location, agenda (at least one must be supplied)
Output Schema	<pre> { "status": "success", "data": { "meeting_id": "uuid", "updated_fields": ["string"], "meeting": { "meeting_id": "uuid", "title": "string", "start_time": "datetime", "end_time": "datetime", "participants": ["string"], "location": "string null", "agenda": "string null", "updated_at": "datetime" } }, "error": null } </pre>
Read/Write Operation	Write
Approval Required	No
Possible Errors	MEETING_NOT_FOUND; NO_FIELDS_PROVIDED; INVALID_TIME_RANGE; DB_WRITE_FAILURE.
Example Call	<pre> { "tool": "UpdateMeeting", "arguments": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "start_time": "2026-07-21T14:00:00", "end_time": "2026-07-21T15:00:00" } } </pre>
Example Result	<pre> { "status": "success", "data": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "updated_fields": ["start_time", "end_time"], "meeting": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "title": "Sprint Planning", "start_time": "2026-07-21T14:00:00", "end_time": "2026-07-21T15:00:00", "participants": ["alice@company.com", "bob@company.com"], "location": "Conference Room B", </pre>

	<pre> "agenda": null, "updated_at": "2026-07-18T09:45:30" }, "error": null } </pre>
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10. Tool: DeleteMeeting

Tool Name	DeleteMeeting
Purpose	Permanently cancels and removes a meeting record. This is a destructive, irreversible operation and requires human approval before execution.
Input Schema	<pre> { "type": "object", "properties": { "meeting_id": { "type": "string", "format": "uuid" }, "reason": { "type": "string", "maxLength": 300 } }, "required": ["meeting_id"] } </pre>
Required Fields	meeting_id
Optional Fields	reason
Output Schema	<pre> { "status": "success", "data": { "meeting_id": "uuid", "deleted": true, "deleted_at": "datetime" }, "error": null } </pre>
Read/Write Operation	Write
Approval Required	Yes — must be routed through ApprovalTool before deletion is committed.
Possible Errors	MEETING_NOT_FOUND; APPROVAL_PENDING; APPROVAL_REJECTED; DB_WRITE_FAILURE.
Example Call	<pre> { "tool": "DeleteMeeting", "arguments": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "reason": "Meeting cancelled by organizer" } } </pre>

Example Result	<pre>{ "status": "success", "data": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "deleted": true, "deleted_at": "2026-07-18T09:50:02" }, "error": null }</pre>
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11. Tool: ExtractMeetingActionItems

Tool Name	ExtractMeetingActionItems
Purpose	Analyzes meeting notes or transcript text supplied for a given meeting and extracts a structured list of action items, optionally converting each into a task draft. Uses the LLM's summarization and extraction capability rather than keyword rules.
Input Schema	<pre>{ "type": "object", "properties": { "meeting_id": { "type": "string", "format": "uuid" }, "transcript": { "type": "string", "minLength": 1, "maxLength": 20000 }, "auto_create_tasks": { "type": "boolean" } }, "required": ["meeting_id", "transcript"] }</pre>
Required Fields	meeting_id, transcript
Optional Fields	auto_create_tasks (defaults to false; if true, each extracted item is passed to CreateTask internally)
Output Schema	<pre>{ "status": "success", "data": { "meeting_id": "uuid", "action_items": [{ "description": "string", "owner": "string null", "due_date": "string null", "created_task_id": "uuid null" }], "item_count": "integer" }, "error": null }</pre>

Read/Write Operation	Write if auto_create_tasks is true; otherwise Read/Derive (analysis-only, no persistent state change beyond an optional extraction log).
Approval Required	No (extraction itself does not require approval; if auto_create_tasks is true, the resulting CreateTask calls follow CreateTask's own approval policy, which is No).
Possible Errors	MEETING_NOT_FOUND; EMPTY_TRANSCRIPT; TRANSCRIPT_TOO_LONG — exceeds 20,000 characters; EXTRACTION_FAILED — LLM returned malformed or unparsable structure; DB_WRITE_FAILURE.
Example Call	<pre>{ "tool": "ExtractMeetingActionItems", "arguments": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "transcript": "Alice will send the updated budget by Friday. Bob needs to review the vendor contract before next week.", "auto_create_tasks": true } }</pre>
Example Result	<pre>{ "status": "success", "data": { "meeting_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "action_items": [{ "description": "Send the updated budget", "owner": "Alice", "due_date": "2026-07-24T17:00:00", "created_task_id": "c3d4e5f6-a7b8-4901-bcde-f23456789012" }, { "description": "Review the vendor contract", "owner": "Bob", "due_date": "2026-07-25T17:00:00", "created_task_id": "d4e5f6a7-b8c9-4012-cdef-345678901234" }], "item_count": 2 }, "error": null }</pre>

12. Tool: ReminderTool

Tool Name	ReminderTool
Purpose	Schedules a reminder notification linked to an existing task or meeting, to be delivered at a specified future time.
Input Schema	<pre>{ "type": "object", "properties": { "linked_entity_type": { "type": "string", "enum": ["task", "meeting"] }, "linked_entity_id": { "type": "string", </pre>

	<pre> "format": "uuid" }, "remind_at": { "type": "string", "format": "date-time" }, "message": { "type": "string", "maxLength": 300 } }, "required": ["linked_entity_type", "linked_entity_id", "remind_at"] } </pre>
Required Fields	linked_entity_type, linked_entity_id, remind_at
Optional Fields	message (defaults to an auto-generated reminder text derived from the linked entity's title)
Output Schema	<pre> { "status": "success", "data": { "reminder_id": "uuid", "linked_entity_type": "string", "linked_entity_id": "uuid", "remind_at": "datetime", "message": "string", "status": "scheduled" }, "error": null } </pre>
Read/Write Operation	Write
Approval Required	No
Possible Errors	ENTITY_NOT_FOUND — referenced task/meeting does not exist; INVALID_REMIND_TIME — remind_at is in the past; INVALID_ENTITY_TYPE; DB_WRITE_FAILURE.
Example Call	<pre> { "tool": "ReminderTool", "arguments": { "linked_entity_type": "meeting", "linked_entity_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "remind_at": "2026-07-21T13:45:00", "message": "Sprint Planning starts in 15 minutes" } } </pre>
Example Result	<pre> { "status": "success", "data": { "reminder_id": "e5f6a7b8-c9d0-4123-defa-456789012345", "linked_entity_type": "meeting", "linked_entity_id": "a1b2c3d4-e5f6-4789-9abc-def012345678", "remind_at": "2026-07-21T13:45:00", "message": "Sprint Planning starts in 15 minutes", "status": "scheduled" }, "error": null } </pre>

	}
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13. Tool: ApprovalTool

Tool Name	ApprovalTool
Purpose	Acts as the mandatory gating mechanism (human-in-the-loop checkpoint) for any tool marked Approval Required: Yes. When such a tool is invoked, LangGraph routes execution to an interrupt state; ApprovalTool presents the pending action to the user and records their decision, which then determines whether the original tool call is committed or discarded.
Input Schema	<pre>{ "type": "object", "properties": { "pending_action_id": { "type": "string", "format": "uuid" }, "target_tool": { "type": "string", "enum": ["DeleteTask", "DeleteMeeting"] }, "decision": { "type": "string", "enum": ["approve", "reject"] }, "comment": { "type": "string", "maxLength": 300 } }, "required": ["pending_action_id", "target_tool", "decision"] }</pre>
Required Fields	pending_action_id, target_tool, decision
Optional Fields	comment
Output Schema	<pre>{ "status": "success", "data": { "pending_action_id": "uuid", "target_tool": "string", "decision": "string", "executed": "boolean", "result": "object null" }, "error": null }</pre>
Read/Write Operation	Write (state transition of the approval workflow; may trigger a downstream Write on the target tool if decision is "approve").

Approval Required	N/A — this tool is the approval mechanism itself and is not gated by another approval step.
Possible Errors	PENDING_ACTION_NOT_FOUND — no such action awaiting approval; ACTION_ALREADY_RESOLVED — decision already recorded; INVALID_DECISION — value not in ["approve","reject"]; DOWNSTREAM_EXECUTION_FAILURE — target tool failed after approval was granted.
Example Call	<pre>{ "tool": "ApprovalTool", "arguments": { "pending_action_id": "f6a7b8c9-d0e1-4234-efab-567890123456", "target_tool": "DeleteTask", "decision": "approve", "comment": "Confirmed duplicate" } }</pre>
Example Result	<pre>{ "status": "success", "data": { "pending_action_id": "f6a7b8c9-d0e1-4234-efab-567890123456", "target_tool": "DeleteTask", "decision": "approve", "executed": true, "result": { "task_id": "6f2a1b2e-4c3d-4a9f-8b2c-1e2f3a4b5c6d", "deleted": true, "deleted_at": "2026-07-18T09:55:47" } }, "error": null }</pre>

14. Summary Matrix

#	Tool Name / Operation / Approval
1	CreateTask — Operation: Write — Approval: No
2	UpdateTask — Operation: Write — Approval: No
3	DeleteTask — Operation: Write — Approval: Yes
4	CompleteTask — Operation: Write — Approval: No
5	ListTasks — Operation: Read — Approval: No
6	SearchTasks — Operation: Read — Approval: No
7	CreateMeeting — Operation: Write — Approval: No
8	UpdateMeeting — Operation: Write — Approval: No
9	DeleteMeeting — Operation: Write — Approval: Yes
10	ExtractMeetingActionItems — Operation: Read / conditional Write — Approval: No (governed by CreateTask policy if auto-create is enabled)
11	ReminderTool — Operation: Write — Approval: No
12	ApprovalTool — Operation: Write (gating) — Approval: N/A (self-referential gate)

15. Error Envelope Convention

All tools that fail return a consistent error envelope, regardless of the specific error code:

```
{
  "status": "error",
  "data": null,
  "error": {
    "code": "ERROR_CODE_STRING",
    "message": "Human-readable description of the failure."
  }
}
```

This uniform structure allows the LangGraph orchestration layer to handle failures generically (e.g., retry, ask the user for clarification, or surface the error) without needing tool-specific exception handling logic at the graph level.

End of Tool Specification Document.